BRIEFCASE

Law Firm Case Management System

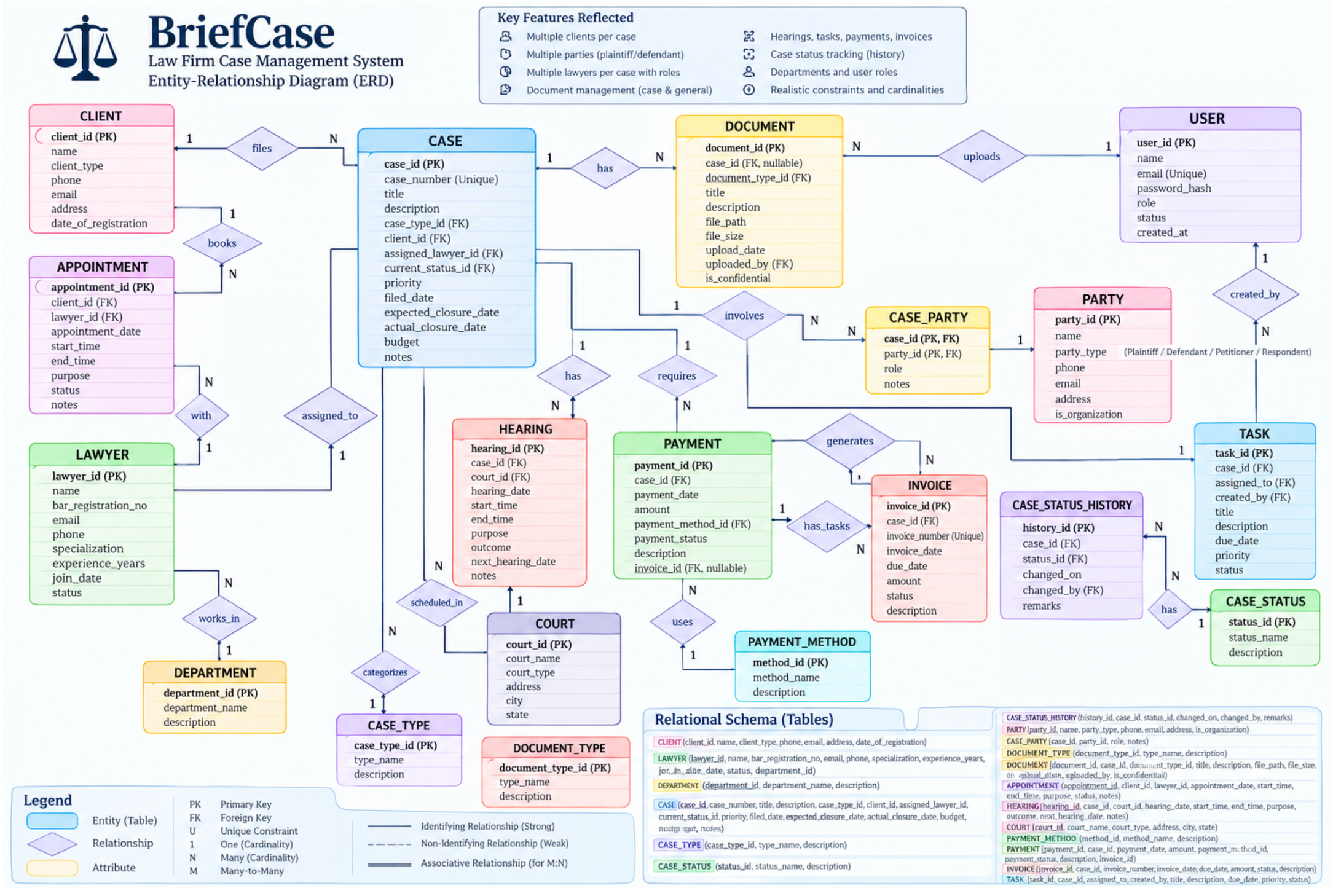
Improved Entity–Relationship Model & Entity Guide

Project Goal Build a realistic, database-driven law-firm platform that can manage clients, cases, lawyers, hearings, documents, tasks, parties and billing.	**Why this model is stronger** It combines core legal case management with workflow, financial management, document management, authentication and audit history.

Core architecture: Client → Case → Lawyers / Parties / Documents / Hearings / Tasks / Payments / Invoices, with status history and user-based audit tracking.

Improved ER Diagram

HD overview of the proposed BriefCase database design. The diagram emphasizes realistic cardinalities, associative entities for M:N relationships, workflow tracking and billing.



Primary entities: CLIENT, CASE, LAWYER, PARTY, DOCUMENT, HEARING, COURT, TASK, PAYMENT and INVOICE. Supporting/reference entities provide categories, statuses, payment methods, departments and document types.

Key Relationships & Business Logic

Relationship	Cardinality	Meaning
CLIENT → CASE	1 : N	One client can have many cases; each case belongs to one client.
CASE ↔ LAWYER	M : N	A case can have multiple lawyers and a lawyer can handle multiple cases; CASE_LAWYER stores role and assignment.
CASE ↔ PARTY	M : N	A case can involve multiple parties; CASE_PARTY stores the party's role.
CASE → DOCUMENT	1 : N	A case can have many documents; each document can be classified by DOCUMENT_TYPE.
CASE → HEARING	1 : N	A case may have many hearings, each scheduled at a court.
CASE → TASK	1 : N	A case can generate multiple tasks assigned to system users.
CASE → PAYMENT	1 : N	A case can have multiple payments.
CASE → INVOICE	1 : N	A case can generate multiple invoices.
CASE → CASE_STATUS_HISTORY	1 : N	Every status change is recorded for auditability.
LAWYER → DEPARTMENT	N : 1	Many lawyers can work within one department.
COURT → HEARING	1 : N	One court can host many hearings.
PAYMENT_METHOD → PAYMENT	1 : N	One payment method can be used for many payments.

Entity Guide

1. CLIENT
Represents a person or organization that hires the law firm.

Key attributes:
• client_id — Primary Key
• name, client_type
• phone, email, address
• date_of_registration

2. CASE
The central entity representing a legal matter handled by the firm.

Key attributes:
• case_id — Primary Key
• case_number — Unique case reference
• title, description
• client_id, case_type_id, current_status_id — Foreign Keys
• priority, filed_date, expected_closure_date, budget

3. LAWYER
Represents lawyers working for the firm and their professional information.

Key attributes:
• lawyer_id — Primary Key
• name, bar_registration_no
• specialization, experience_years
• department_id — Foreign Key
• status

4. CASE_LAWYER
Associative entity resolving the many-to-many relationship between CASE and LAWYER.

Key attributes:
• case_id + lawyer_id — Composite Primary Key
• role — Lead Lawyer / Associate / Reviewer
• assigned_date
• is_primary

Entity Guide

5. DEPARTMENT
Represents a legal department or practice area within the firm.

Key attributes:
• department_id — Primary Key
• department_name
• description

6. APPOINTMENT
Represents meetings between clients and lawyers.

Key attributes:
• appointment_id — Primary Key
• client_id, lawyer_id — Foreign Keys
• appointment_date, start_time, end_time
• purpose, status, notes

7. PARTY
Represents people or organizations involved in a case, such as plaintiffs or defendants.

Key attributes:
• party_id — Primary Key
• name, party_type
• phone, email, address
• is_organization

8. CASE_PARTY
Associative entity connecting CASE and PARTY, allowing multiple parties per case.

Key attributes:
• case_id + party_id — Composite Primary Key
• role — Plaintiff / Defendant / Petitioner / Respondent
• notes

Entity Guide

9. CASE_TYPE
Categorizes legal cases.

Key attributes:
• case_type_id — Primary Key
• type_name — Criminal / Civil / Corporate / etc.
• description

10. DOCUMENT
Stores documents/files associated with a case.

Key attributes:
• document_id — Primary Key
• case_id, document_type_id — Foreign Keys
• title, description
• file_path, file_size, upload_date
• uploaded_by, is_confidential

11. DOCUMENT_TYPE
Categorizes uploaded legal documents.

Key attributes:
• document_type_id — Primary Key
• type_name — Evidence / Petition / Contract / etc.
• description

12. USER
Represents authenticated system users such as administrators, lawyers, assistants and staff.

Key attributes:
• user_id — Primary Key
• name, email — Unique
• password_hash
• role, status, created_at

Entity Guide

13. HEARING
Represents a court hearing scheduled for a case.

Key attributes:
• hearing_id — Primary Key
• case_id, court_id — Foreign Keys
• hearing_date, start_time, end_time
• purpose, outcome, next_hearing_date, notes

14. COURT
Represents the court or tribunal where a hearing takes place.

Key attributes:
• court_id — Primary Key
• court_name, court_type
• address, city, state

15. PAYMENT
Records money paid toward a case or invoice.

Key attributes:
• payment_id — Primary Key
• case_id, payment_method_id — Foreign Keys
• payment_date, amount
• payment_status, description, invoice_id

16. PAYMENT_METHOD
Stores supported payment methods.

Key attributes:
• method_id — Primary Key
• method_name — UPI / Cash / Card / Bank Transfer
• description

Entity Guide

17. INVOICE
Represents a bill generated for legal services.

Key attributes:
• invoice_id — Primary Key
• case_id — Foreign Key
• invoice_number — Unique
• invoice_date, due_date, amount, status, description

18. CASE_STATUS
Stores the possible lifecycle states of a case.

Key attributes:
• status_id — Primary Key
• status_name — New / Filed / Hearing / Settled / Closed / etc.
• description

19. CASE_STATUS_HISTORY
Maintains the complete audit trail of case-status changes.

Key attributes:
• history_id — Primary Key
• case_id, status_id, changed_by — Foreign Keys
• changed_on
• remarks

20. TASK
Represents work that must be completed for a case.

Key attributes:
• task_id — Primary Key
• case_id, assigned_to, created_by — Foreign Keys
• title, description, due_date
• priority, status

Recommended Real-World Application Modules

Client Portal	Client registration, profile, case overview, appointments and secure document access.
Lawyer Dashboard	Assigned cases, upcoming hearings, tasks, documents and case activity.
Case Management	Case creation, case type, lawyers, parties, documents, hearings and status tracking.
Task & Workflow	Assign tasks, deadlines, priorities, progress and completion tracking.
Document Management	Upload, classify, search and control confidential case documents.
Hearing Calendar	Court-wise hearing schedule, reminders, outcomes and next hearing dates.
Billing	Invoices, payments, payment methods, pending dues and payment history.
Administration	User roles, departments, case statuses, document types and audit history.

Suggested project positioning: Present BriefCase as a complete Legal Case Management System rather than only a CRUD database. The ER model supports a practical web/mobile application with role-based access, case workflow, document handling, hearing management, task assignment and billing.